

SOP for SMS-ICU score validation

1. Filter the adult ICUs that collected data during the study period from **1st of January, 2023** to **31st of December, 2023**. Using them, please populate table 1 in **SMS-ICU metadata and results document.docx** document
2. From the adult ICUs that collected data during study period, the eligibility of participating ICUs should be determined by the criterion for ICU inclusion as defined below,
 - Adult ICUs which started data collection on or prior to **1st of January, 2022** (allowing a lag period of at least one year to maximise data quality)
3. Once the eligible ICUs are determined, from the eligible ICUs, patients admitted between **1st of January 2023** and **31st of December, 2023** should be extracted
4. Extracted patients' data should be mapped according to the attached [data dictionary](#)
5. Before running the shared scripts, please ensure that the following R libraries are installed. If they are not, try installing them using the commands provided in the brackets:
 - **ggplot2** [install.packages("ggplot2")]
 - **pROC** [install.packages("pROC")]
 - **CalibrationCurves** [install.packages("CalibrationCurves")]
 - **DescTools** [install.packages("DescTools")]
 - **glue** [install.packages("glue")]
 - **dplyr** [install.packages("dplyr")]
 - **readr** [install.packages("readr")]
 - **ems** [install.packages("ems")]
6. To validate mapping, run the script **0_data_mapping_validation.R** (please follow the instructions in the script)
7. After mapping the data to the required format, to validate the in-hospital mortality, run the script **1_hospital_mortality_validation.R** (please follow the instructions in the script)

It will first apply the following inclusion and exclusion criteria to the patients,

- Patient encounters of age < 18 years
- Patient encounters of readmission
- Patient encounters transferred to another hospital
- Patient encounters with missing survival status for in-hospital

Once the criteria is applied, it will generate a csv file named

pop_desc_for_hospital_mortality. Using that csv, please populate table 2 in **SMS-ICU metadata and results document.docx** document. Kindly note that following rows will not be in the csv and they need to be populated as well,

- Number of ICUs included in the study
- Number of male patients (% of total)
- Number of included patients receiving NIV/CPAP* (% of total)
- Admission diagnosis (APACHE classification*)

It will identify complete cases where all components required to calculate the SMS-ICU score are available and will also apply normal imputation for missing data

It will then calculate both the SMS-ICU score and the probability of death

Then it will run the validation on following patient groups (with and without imputation) and generate respective figures and csvs,

<i>Patient group</i>	<i>Results that the script produces</i>
All patient encounters	roc_curve_hospital_mortality_all.png calibration_curve_hospital_mortality_all.png results_for_hospital_mortality_all.csv roc_curve_hospital_mortality_all(complete_case).png calibration_curve_hospital_mortality_all(complete_case).png results_for_hospital_mortality_all(complete_case).csv
Medical admissions	roc_curve_hospital_mortality_medical.png calibration_curve_hospital_mortality_medical.png results_for_hospital_mortality_medical.csv roc_curve_hospital_mortality_medical(complete_case).png calibration_curve_hospital_mortality_medical(complete_case).png results_for_hospital_mortality_medical(complete_case).csv
Emergency surgery admissions	roc_curve_hospital_mortality_emergency_surgery.png calibration_curve_hospital_mortality_emergency_surgery.png results_for_hospital_mortality_emergency_surgery.csv roc_curve_hospital_mortality_emergency_surgery(complete_case).png calibration_curve_hospital_mortality_emergency_surgery(complete_case).png results_for_hospital_mortality_emergency_surgery(complete_case).csv
Scheduled surgery admissions	roc_curve_hospital_mortality_elective_surgery.png calibration_curve_hospital_mortality_elective_surgery.png results_for_hospital_mortality_elective_surgery.csv roc_curve_hospital_mortality_elective_surgery(complete_case).png calibration_curve_hospital_mortality_elective_surgery(complete_case).png results_for_hospital_mortality_elective_surgery(complete_case).csv
Admissions with invasive mechanical ventilation	roc_curve_hospital_mortality_invasive.png calibration_curve_hospital_mortality_invasive.png results_for_hospital_mortality_invasive.csv roc_curve_hospital_mortality_invasive(complete_case).png calibration_curve_hospital_mortality_invasive(complete_case).png results_for_hospital_mortality_invasive(complete_case).csv
Admissions after excluding inter-hospital transfers	roc_curve_hospital_mortality_inter_hos_transfers_excluded.png calibration_curve_hospital_mortality_inter_hos_transfers_excluded.png results_for_hospital_mortality_inter_hos_transfers_excluded.csv roc_curve_hospital_mortality_inter_hos_transfers_excluded(complete_case).png calibration_curve_hospital_mortality_inter_hos_transfers_excluded(complete_case).png results_for_hospital_mortality_inter_hos_transfers_excluded(complete_case).csv

8. Then, to validate the ICU mortality, run the script **2_icu_mortality_validation.R** (please follow the instructions in the script)

It will apply the following inclusion and exclusion criteria to the patients

- Patient encounters of age < 18 years
- Patient encounters of readmission
- Patient encounters transferred to another hospital
- Patient encounters with missing survival status for ICU

Once the criteria is applied, it will generate a csv file named **pop_desc_for_icu_mortality**. Using that csv, please populate table 3 in **SMS-ICU metadata and results document.docx** document. Kindly note that following rows will not be in the csv and they need to be populated as well,

- Number of ICUs included in the study
- Number of male patients (% of total)
- Number of included patients receiving NIV/CPAP* (% of total)
- Admission diagnosis (APACHE classification*)

It will identify complete cases where all components required to calculate the SMS-ICU score are available and will also apply normal imputation for missing data

It will then calculate both the SMS-ICU score and the probability of death

Then it will run the validation on following patient groups (with and without imputation) and generate respective figures and csvs,

Patient group	Results that the script produces
All patient encounters	roc_curve_icu_mortality_all.png calibration_curve_icu_mortality_all.png results_for_icu_mortality_all.csv roc_curve_icu_mortality_all(complete_case).png calibration_curve_icu_mortality_all(complete_case).png results_for_icu_mortality_all(complete_case).csv
Medical admissions	roc_curve_icu_mortality_medical.png calibration_curve_icu_mortality_medical.png results_for_icu_mortality_medical.csv roc_curve_icu_mortality_medical(complete_case).png calibration_curve_icu_mortality_medical(complete_case).png results_for_icu_mortality_medical(complete_case).csv
Emergency surgery admissions	roc_curve_icu_mortality_emergency_surgery.png calibration_curve_icu_mortality_emergency_surgery.png results_for_icu_mortality_emergency_surgery.csv roc_curve_icu_mortality_emergency_surgery(complete_case).png calibration_curve_icu_mortality_emergency_surgery(complete_case).png results_for_icu_mortality_emergency_surgery(complete_case).csv
Scheduled surgery admissions	roc_curve_icu_mortality_elective_surgery.png calibration_curve_icu_mortality_elective_surgery.png results_for_icu_mortality_elective_surgery.csv roc_curve_icu_mortality_elective_surgery(complete_case).png calibration_curve_icu_mortality_elective_surgery(complete_case).png

	results_for_icu_mortality_elective_surgery(complete_case).csv
Admissions with invasive mechanical ventilation	roc_curve_icu_mortality_invasive.png calibration_curve_icu_mortality_invasive.png results_for_icu_mortality_invasive.csv roc_curve_icu_mortality_invasive(complete_case).png calibration_curve_icu_mortality_invasive(complete_case).png results_for_icu_mortality_invasive(complete_case).csv
Admissions after excluding inter-hospital transfers	roc_curve_icu_mortality_inter_hos_transfers_excluded.png calibration_curve_icu_mortality_inter_hos_transfers_excluded.png results_for_icu_mortality_inter_hos_transfers_excluded.csv roc_curve_icu_mortality_inter_hos_transfers_excluded(complete_case). png calibration_curve_icu_mortality_inter_hos_transfers_excluded(complete _case).png results_for_icu_mortality_inter_hos_transfers_excluded(complete_case). csv

9. In the end, populated ***SMS-ICU metadata and results document.docx*** document along with 26 csv files and 48 png files (figures) need to be submitted